

Proposed Prescribing Information
For the use of a Registered Medical Practitioner or a Hospital or a Laboratory

**TORIDIP 10 MG TABLETS AND TORIDIP 20 MG TABLETS
LERCANIDIPINE HYDROCHLORIDE TABLETS**

BRAND OR PRODUCT NAME

TORIDIP 10 mg Tablets

TORIDIP 20 mg Tablets

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

TORIDIP 10 mg Tablets

Each film coated tablet contains:

Lercanidipine Hydrochloride.....10mg

equivalent to Lercanidipine.....9.4 mg

Colours: Yellow oxide of Iron & Titanium dioxide

TORIDIP 20 mg Tablets

Each film coated tablet contains:

Lercanidipine Hydrochloride.....20mg

equivalent to Lercanidipine18.8 mg

Colours: Red oxide of Iron & Titanium dioxide

PRODUCT DESCRIPTION

TORIDIP 10 mg Tablets

Yellow coloured round, shaped biconvex, film coated tablets with break-line on one side and plain on other side.

TORIDIP 20 mg Tablets

Pink coloured round, shaped biconvex, film coated tablets with break-line on one side and plain on other side.

DOSAGE FORM

Film coated tablet

PHARMACODYNAMICS / PHARMACOKINETICS

Pharmacodynamics

Selective calcium channel blockers with mainly vascular effects

ATC code: C08CA13

Lercanidipine is a calcium antagonist of the dihydropyridine group and inhibits the transmembrane influx of calcium into cardiac and smooth muscle. The mechanism of its antihypertensive action is due to a direct relaxant effect on vascular smooth muscle thus lowering total peripheral resistance. Despite its short pharmacokinetic plasma half-life, lercanidipine is endowed with a prolonged antihypertensive activity because of its high membrane partition coefficient, and is devoid of negative inotropic effects due to its high vascular selectivity. Since the vasodilatation induced by Lercanidipine is gradual in onset, acute hypotension with reflex tachycardia has rarely been observed in hypertensive patients. As for other asymmetric 1,4-dihydropyridines, the antihypertensive activity of lercanidipine is mainly due to its (S)-enantiomer.

Pharmacokinetics

Lercanidipine is completely absorbed from the gastrointestinal tract after oral doses but undergoes extensive saturable first-pass metabolism. Bioavailability is low but is increased in the presence of food. Peak plasma concentrations occur about 1.5 to 3 hours after oral dosage. Lercanidipine is rapidly and widely distributed. It is more than 98% bound to plasma proteins. Lercanidipine is extensively metabolised, primarily by the cytochrome P450 isoenzyme CYP3A4, mainly to inactive metabolites; about 50% of an oral dose is excreted in the urine. A terminal elimination half-life of about 2 to 5 hours has been reported, but studies using a more sensitive assay have suggested a value of 8 to 10 hours

INDICATION

Lercanidipine is indicated for the treatment of mild to moderate essential hypertension.

RECOMMENDED DOSAGE AND MODE OF ADMINISTRATION

The recommended dosage is 10 mg orally once a day at least 15 minutes before meals; the dose may be increased to 20 mg depending on the individual patient's response. Dose titration should be gradual, because it may take about 2 weeks before the maximal antihypertensive effect is apparent. Some individuals, not adequately controlled on a single antihypertensive agent, may benefit from the addition of Lercanidipine to therapy with a beta-adrenoceptor blocking drug (atenolol), a diuretic (hydrochlorothiazide) or an angiotensin converting enzyme inhibitor (captopril or enalapril). Since the dose-response curve is steep with a plateau at doses between 20-30 mg, it is unlikely that efficacy will be improved by higher doses; whereas side effects may increase.

Use in the elderly: although the pharmacokinetic data and clinical experience suggest that no adjustment of the daily dosage is required, special care should be exercised when initiating treatment in the elderly.

Use in children: since there is no clinical experience in patients under the age of 18 years, use in children is not currently recommended.

Use in renal or hepatic dysfunction: special care should be exercised when treatment is commenced in patients with mild to moderate renal or hepatic dysfunction. Although the usually recommended dose schedule may be tolerated by these subgroups, an increase in dose to 20 mg daily must be approached with caution. The antihypertensive effect may be enhanced in patients with hepatic impairment and consequently an adjustment of the dosage should be considered. Lercanidipine is not recommended for use in patients with severe hepatic impairment or in patients with severe renal impairment (GFR < 30 ml/min).

CONTRAINDICATIONS

Hypersensitivity to the active substance "lercanidipine", to any dihydropyridine or to any of the excipients of the medicinal product.

- Pregnancy and lactation.
- Women of child-bearing potential unless effective contraception is used.
- Left ventricular outflow tract obstruction.
- Untreated congestive cardiac failure.
- Unstable angina pectoris.
- Severe renal or hepatic impairment.
- Within 1 month of a myocardial infarction.
- Co-administration with:
 - strong inhibitors of CYP3A4,
 - cyclosporin,
 - grapefruit juice.

WARNINGS AND PRECAUTIONS

Special care should be exercised when Lercanidipine is used in patients with sick sinus syndrome (if a pacemaker is not in situ). Although no impairment of ventricular function reported, care is also required in patients with LV dysfunction. It has been suggested that some short-acting dihydropyridines may be associated with increased cardiovascular risk in patients with ischaemic heart disease. Although Lercanidipine is long-acting caution is required in such patients.

Some dihydropyridines may rarely lead to precordial pain or angina pectoris. Very rarely patients with pre-existing angina pectoris may experience increased frequency, duration or severity of these attacks. Isolated cases of myocardial infarction may be observed.

Use in renal or hepatic dysfunction: special care should be exercised when treatment is commenced in patients with mild to moderate renal or hepatic dysfunction. Although the usually recommended dose schedule may be tolerated by these subgroups, an increase in dose to 20 mg daily must be approached with caution. The antihypertensive effect may be enhanced in patients with hepatic impairment and consequently an adjustment of the dosage should be considered.

Lercanidipine is not recommended for use in patients with severe hepatic impairment or in patients with severe renal impairment (GFR < 30 ml/min). Alcohol should be avoided since it may potentiate the effect of vasodilating antihypertensive drugs.

Inducers of CYP3A4 like anticonvulsants (e.g. phenytoin, carbamazepine) and rifampicin may reduce lercanidipine's plasma levels and therefore the efficacy of lercanidipine may be less than expected.

INTERACTIONS WITH OTHER MEDICAMENTS

Lercanidipine is known to be metabolised by the CYP3A4 enzyme and, therefore, inhibitors and inducers of CYP3A4 administered concurrently may interact with the metabolism and elimination of lercanidipine.

Co-prescription of Lercanidipine with inhibitors of CYP3A4 (e.g. ketoconazole, itraconazole, ritonavir, erythromycin, troleandomycin) should be avoided. A strong CYP3A4 inhibitor, ketoconazole, considerably increases plasma levels of lercanidipine.

Cyclosporin and lercanidipine should not be administered together.

Increased plasma levels of both lercanidipine and cyclosporin have been reported following concomitant administration.

Lercanidipine should not be taken with grapefruit juice.

As for other dihydropyridines, lercanidipine is sensitive to inhibition of metabolism by grapefruit juice, with a consequent rise in its systemic availability and increased hypotensive effect.

Concomitantly administration of oral midazolam at a dose of 20 mg with lercanidipine reported to increase lercanidipine's absorption and decrease the rate of absorption.

Caution should be exercised when Lercanidipine is co-prescribed with other substrates of CYP3A4, like terfenadine, astemizole, class III antiarrhythmic drugs such as amiodarone, quinidine.

Co-administration of Lercanidipine with CYP3A4 inducers like anticonvulsants (e.g. phenytoin, carbamazepine) and rifampicin should be approached with caution since the antihypertensive effect may be reduced and blood pressure should be monitored more frequently than usual.

Upon co-administration Lercanidipine with metoprolol, a β -blocker eliminated mainly by the liver, the bioavailability of metoprolol was not reported to change while that of lercanidipine was reported to be reduced. This effect may be due to the reduction in the hepatic blood flow caused by β -blockers and may therefore occur with other drugs of this class.

Consequently, lercanidipine may be safely administered with beta-adrenoceptor blocking drugs, but dose adjustment may be required.

Fluoxetine (an inhibitor of CYP2D6 and CYP3A4), has no clinically relevant modification of the pharmacokinetics of lercanidipine.

Concomitant administration of cimetidine 800 mg daily does not cause significant modifications in plasma levels of lercanidipine, but at higher doses caution is required since the bioavailability and the hypotensive effect of lercanidipine may be increased.

Co-administration of 20 mg lercanidipine with β -methyldigoxin reported no evidence of pharmacokinetic interaction. Digoxin reported to increased absorption following dosing with 20 mg lercanidipine. Patients on concomitant digoxin treatment should be closely monitored clinically for signs of digoxin toxicity.

No interaction is expected when lercanidipine is administered in the morning and simvastatin in the evening, as indicated for such drug.

The co-administration of 20 mg lercanidipine not reported to alter the pharmacokinetics of warfarin. Alcohol should be avoided since it may potentiate the effect of vasodilating antihypertensive drugs.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

There are no reports of a teratogenic effect in the rat and the rabbit and reproductive performance in the rat was unimpaired. Nevertheless, since there is no clinical experience with lercanidipine in pregnancy and lactation, and other dihydropyridine compounds have been found teratogenic in animals, Lercanidipine should not be administered during pregnancy or to women with child-bearing potential unless effective contraception is used. Because of high lipophilicity of lercanidipine, distribution in milk may be expected. Therefore, it should not be administered to nursing mothers.

ADVERSE EFFECTS / UNDESIRABLE EFFECTS

The following are undesirable effects with lercanidipine:

Immune System Disorders: hypersensitivity

Psychiatric Disorders: somnolence

Nervous System Disorders: headache; dizziness

Cardiac Disorders: angina pectoris, tachycardia; palpitations

Vascular Disorders: flushing, syncope

Gastrointestinal Disorders: nausea; dyspepsia; diarrhoea; abdominal pain; vomiting

Skin and Subcutaneous Tissue Disorders: rash

Musculoskeletal, Connective Tissue and Bone Disorders: myalgia

Renal and Urinary Disorders: polyuria

General Disorders and Administration Site Conditions: oedema peripheral, asthenia; fatigue

The following undesirable effects were reported very rarely: gingival hypertrophy, reversible increases in serum levels of hepatic transaminases, hypotension, urinary frequency and chest pain.

Some dihydropyridines may rarely lead to precordial pain or angina pectoris. Lercanidipine does not appear to influence adversely blood sugar or serum lipid levels.

OVERDOSE AND TREATMENT

Three cases of overdose were reported (150 mg, 280 mg and 800 mg of lercanidipine, respectively, ingested in an attempt to commit suicide).

<i>Dose level</i>	<i>Signs/Symptoms</i>	<i>Management</i>	<i>Outcome</i>
150 mg + undefined amount of alcohol	Sleepiness	Gastric lavage Active charcoal	Recovered
280 mg + 5.6 mg moxonidine	Cardiogenic shock Severe myocardial ischaemia Mild renal failure	High-dose catecholamines Furosemide Digitalis Parenteral plasma expanders	Recovered
800 mg	Emesis Hypotension	Active charcoal Cathartics Dopamine i.v.	Recovered

As with other dihydropyridines, overdosage might be expected to cause excessive peripheral vasodilatation with marked hypotension and reflex tachycardia. In case of severe hypotension, bradycardia and unconsciousness, cardiovascular support could be helpful, with intravenous atropine for bradycardia.

In view of the prolonged pharmacological effect of lercanidipine, it is essential that the cardiovascular status of patients who take an overdose is monitored for 24 hours at least. There is no information on the value of dialysis. Since the drug is highly lipophilic, it is most probable that plasma levels are no guide to the duration of the period of risk and dialysis may not be effective.

STORAGE CONDITIONS

Store below 30° C.

PACKAGING AVAILABLE

Lercanidipine Hydrochloride Tablets are packed in White opaque PVC/PVDC-Alu blister of 14 Tablets. Such blister containing 14 Tablets are packed in boxes of 28's along with the package insert.

DATE OF REVISION OF PACKAGE INSERT

01/04/2014

NAME AND ADDRESS OF MANUFACTURER



TORRENT PHARMACEUTICALS LTD.

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INDIA.